

[Co-1] Assessment Tool-2

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Slot — B

Code — MLA0401

Course — Deep Learning

47
50

marks
94%

Introduction -

Neural Network is a machine learning model that learns pattern from data and predict output.

During training - model learns

During testing - predicts result

$$\text{Training MSE} = 0.85$$

$$\text{Testing MSE} = 5.20$$

MSE (Mean Squared Error)

$$\text{MSE} = \frac{1}{n} \sum (\text{Actual} - \text{Predicted})^2$$

Lower MSE - better Predicted.

Higher MSE - large Predicted errors.

Given,

$$\text{Training MSE} = 0.85$$

$$\text{Testing MSE} = 5.20$$

$$5.20 > 0.85$$

the model has high testing error

$\therefore$ It is overfitting.

Reasons for High Testing Error

- Overfitting
- Small Training Dataset
- complex Neural Network
- Noise in data
- Lack of Regularization
- Improper data splitting.

corrective Measures

- Increase Training data
- use Regularization
- Early stopping
- simplify the Model
- Remove Noise
- Tune Hyperparameters

Training error	Testing error	Interpretation
High	High	underfitting
Low	High	overfitting
Low	Low	Good Model

this model is overfitting.

conclusion

since the testing error is much larger than the training error, the model is suffering from overfitting.